

Major Project

Proposal Form

Complete this form before your proposal presentation and have it to hand during the session. Upload the completed form to Learning Space alongside your presentation slides. Your supervisor will add their feedback at or following your first supervision session after the presentation. The form is a record of the key commitments made in your proposal; it is not a substitute for the presentation itself.

1. Student details

Full name	Morgan Tomos Hodge
Student number	2510355
Programme	MA Game Design
Supervisor	Doctor Jeff Howard
Date of proposal presentation	24/06/2026

2. Project title and research question

Proposed project title	Environmental Guidance in Digital Games: Influencing Player Navigation Without Explicit Instruction
Research question	Which environmental design techniques are most effective at influencing player navigation and decision making without explicit guidance systems?

3. Artefact

Artefact type	Software product / Blog
---------------	-------------------------

Artefact description and intended audience

The primary artefact will be a series of playable prototypes developed in unreal engine that investigate different environmental guidance techniques. Supporting this artefact will be a research blog that documents findings from the literature review, contextual review, development process, and user testing.

More specifically I will be creating 4 small playable prototypes within Unreal Engine, each of these small prototypes will demonstrate a different method of environmental guidance. This will include Lighting, Architectural, Landmark, Audio, and combined guidance techniques.

My primary audience is myself and potential future employers. I intend to use this project as a key portfolio piece, as it demonstrates my approach to game design and reflects my passion for this discipline. I will be applying for roles in Technical Design, Level Design, and Systems Design, and I believe this project provides opportunities to showcase skills relevant to all three areas.

This project is particularly relevant to studios that develop narrative-driven, immersive, or exploration focused experiences where environmental guidance forms an important part of the player experience. Examples include AAA studios such as Naughty Dog, Arkane Studios, Ubisoft, and PlayStation Studios, as well as independent developers creating first person exploration, action adventure, and immersive simulation games. The project demonstrates skill in level design, player guidance, prototyping, user testing, and design research, all of which are valuable within these roles. As a result, it has the potential to become a strong and versatile addition to my professional portfolio.

A secondary audience is other game designers, particularly level designers. The findings from this project may help inform their own approaches to player guidance and environmental navigation. Additionally, indie studios may benefit from this research, as smaller development teams often lack dedicated designers.

4. Field context and key references

Field context

Summarise the state of the art relevant to your project. What existing work, research, or practice does your project build upon or respond to?

Player guidance is a fundamental aspect of game design, as designers must communicate objectives and direct players through virtual environments. Traditional methods of guidance often rely on explicit systems such as quest markers, minimaps, objective indicators, and tutorial prompts. While effective, these systems can reduce immersion and limit opportunities for player driven discovery.

Research into wayfinding and navigation suggests that environmental factors such as landmarks, spatial layout, visual cues, and environmental composition play a significant role in how individuals navigate unfamiliar spaces. Carlson (2010) argues that users create cognitive maps from the environmental information, while Abdullahi (2025) found that environmental configuration can influence navigation behaviour and route selection. These findings suggest that environmental design itself can be used as a method of player guidance.

This concept is also widely discussed within level design literature and professional practice. Totten (2019) argues that architectural composition and spatial organisation can be used to communicate information and direct player movement through virtual environments. Professional level designers frequently discuss the use of lighting, landmarks, framing, environmental storytelling, and visual hierarchy as methods of subtly directing player attention and movement.

A notable example of environmental guidance can be found in the concept of the Disney “Weenie”, a term used to describe a prominent visual landmark designed to attract attention and draw visitors through a

space. This principle has influenced level design practice and demonstrates how environmental features can guide movement without explicit instruction.

More recently, player guidance has become a topic of discussion within the game design community due to the increasing use of highly visible environmental cues often referred to by players as the “yellow paint” approach. Games such as the recent entries in the Tomb Raider, Resident Evil, and Final Fantasy series have used yellow paint or similar contrasting visual markers to highlight interactable objects and recommended routes the player should follow. While these techniques serve their purpose and help players navigate and play the game, some users are strongly against it. They believe this reduces immersion and makes environmental navigation feel overly directed. This debate highlights the ongoing challenge faced by designers: balancing clear player guidance with exploration, immersion, and player agency.

This project will build upon existing research and industry practices by investigating which environmental design techniques are most effective at influencing player navigation and decision making without explicit guidance systems. Through the development and testing of playable prototypes, the project aims to contribute practical recommendations for game designers seeking to create more immersive and intuitive player experiences.

Key references

List a minimum of three key academic or professional sources that inform your project. Use an appropriate citation convention consistently.

Books

- Schell, J. (2024) *The Art of Game Design: A Book of Lenses*. 4th edn. Boca Raton: CRC Press. Available at: <https://schellgames.com/art-of-game-design>
- Salen, K. and Zimmerman, E. (2004) *Rules of Play: Game Design Fundamentals*. Cambridge, MA: MIT Press. Available at: <https://mitpress.mit.edu/9780262240451/rules-of-play/>
- Totten, C.W. (2019) *An Architectural Approach to Level Design*. 2nd edn. Boca Raton: CRC Press. Available at: <https://www.amazon.co.uk/Architectural-Approach-Level-Design/dp/1466585412>

Academic Articles

- Carlson, L.A., Hölscher, C., Shipley, T.F. and Dalton, R.C. (2010) ‘Getting Lost in Buildings’, *Current Directions in Psychological Science*, 19(5), pp. 284–289. Available at: <https://journals.sagepub.com/doi/abs/10.1177/0963721410383243>
- Abdullahi, M.S., Ryu, J.J., Kim, S.-N. and Lee, J. (2025) ‘The influence of urban configuration on wayfinding propensity: A video game-based study’, *Urban Design International*. Available at: <https://doi.org/10.1057/s41289-025-00290-7>
- Nacke, L.E. and Deterding, S. (2017) ‘The maturing of gamification research’, *Computers in Human Behavior*, 71, pp. 450–454. Available at: https://eprints.whiterose.ac.uk/id/eprint/111254/1/Deterding_Nacke_Editorial_Maturing_Gamification_Research.pdf
- Justinmind (2025) ‘Game UI design: The mechanics of fun experiences’. *Justinmind Blog*, 7 February. Available at: [Justinmind – Game UI design: The mechanics of fun experiences](https://www.justinmind.com/blog/game-ui-design-the-mechanics-of-fun-experiences/)

- Wirtz, B. (2025) 'Yellow Paint in Game Design: What's the Problem?', *Game Designing*. Available at: <https://gamedesigning.org/learn/yellow-paint/>

Videos

- Game Developers Conference (2010) *Ten Principles for Good Level Design*. Presented by Dan Taylor. Available at: <https://www.youtube.com/watch?v=iNEe3KhMvXM>
- Game Developers Conference (2017) *Level Design Workshop: Tricks of the Trade*. Presented by Steve Lee. Available at: <https://www.gamedeveloper.com/design/gdc-2017-level-design-workshop>
- Blundell, J. (Various interviews) Discussion of player flow, environmental storytelling and navigation design in Call of Duty Zombies. Available at: <https://www.youtube.com/watch?v=i7Cvu4Aq3IQ>
- What is a Disney Weenie? Available at: <https://www.youtube.com/watch?v=S7COKSbmXWk>

Games (All these games use environmental guidance incredibly well without constantly relying on UI markers)

- Portal (Valve, 2007)
- The Last of Us (Naughty Dog, 2013)
- BioShock (2K Boston/2K Australia, 2007)

5. Methodology

Research and development approach

How will you investigate your research question? Describe the methods, tools, and processes you will use. Justify your choices in relation to your artefact type and research question.

I will investigate my research question through a combination of literature review, contextual analysis, prototype development, and user testing.

The project will begin with a thorough review of academic literature relating to player navigation, wayfinding, environmental storytelling, and level design. This research will be supported by an analysis of industry sources, including GDC presentations, developer talks, and professional articles. The aim of this research phase is to identify current best practices and establish an existing foundation for the project.

The findings from the literature and contextual review will influence the development of a series of small-scale playable prototypes created in unreal engine 5. Each prototype will explore a different environmental guidance technique, such as lighting, landmarks, and audio cues. Rather than investigating whether guidance techniques should be combined, the project will also explore how these techniques can be combined effectively to influence player navigation without relying on explicit guidance systems such as objective markers or yellow paint.

User testing will be conducted during the later stages of development to evaluate the effectiveness of each environmental guidance technique. Participants will be asked to complete a navigation task within each prototype while their behaviour is observed and recorded. Quantitative data such as completion time, route selection, and instances where participants become lost or deviate from the intended path will be collected. Following each playtest session, participants will complete a short questionnaire to gather qualitative feedback regarding the clarity, immersion, and perceived effectiveness of the guidance techniques. The findings from these tests will be compared across the prototypes to identify which

environmental guidance methods are most effective at influencing player navigation and decision making without relying on explicit guidance systems.

This methodology is appropriate because the research question focuses on player navigation and environmental guidance, which are best investigated through the creation of playable prototypes. User testing will allow the effectiveness of different guidance techniques to be evaluated in practice, while the lit review will ensure the designs are informed by existing research and industry practice. This approach will generate both practical artefacts and evidence-based findings relevant to game design.

6. Ethics

Full ethics application
required?

Yes – Low/Medium Risk

Ethics considerations

Identify any legal, social, ethical, or professional issues associated with your project. If a full ethics application is required, note its current status. If not, briefly explain why no significant ethical risks are present.

Ethical Considerations

Accessibility Considerations

- The prototypes should consider accessibility where possible
- User testing should not exclude participants unnecessarily
- Instructions and questionnaires should be clear and understandable

Playtesting and Participant Consent

- Participants will be informed about the purpose of the testing before taking part
- Participation will be voluntary
- Participants will be able to withdraw at any time
- Feedback will be anonymous when analysed and reported

Research Integrity

- All academic, industry, and online sources will be properly referenced to avoid plagiarism
- Research findings will be reported accurately and without intentionally manipulating results

Legal Considerations

GDPR/Data Protection Act 2018

- Any data collected during playtesting (questionnaires, observations, recordings) will be stored securely
- Personal information will not be shared with third parties
- Any collected data will be anonymised before analysis when appropriate

Intellectual Property and Asset licensing

- Any third-party assets used within the prototypes will be sourced from legitimate providers such as FAB or other licensed asset marketplaces
- Asset licences will be reviewed to ensure they permit educational and portfolio use
- Appropriate attribution will be provided where required by licence agreements

Copyright

- Academic articles, books, images, videos, and other research materials will be properly cited and referenced
- No copyright work will be used without permission or appropriate licencing

7. Project plan

Complete the table below with your key milestones and any anticipated blockers. You are not expected to fill every row — use as many as are meaningful for your project.

Week	Key milestone or deliverable	Potential blocker / contingency
Week 2 8.06.2026	Complete literature review and identify key theories and resources I will use. Finalise research question and proposal form	Need to be 100% sure this is the project I want to pursue as I will be spending a lot of time on this and it is my major portfolio piece
Week 3 Proposal due 24.06	Polish and submit proposal. Viva Presentation	Work shifts have been occupying a lot of my time, just returning from Falmouth
Week 4 29.06.2026	Analyse case study games and document examples of environmental guidance techniques. Produce initial design concepts for prototypes.	
Week 5 6.07.2026	Create a reusable scene setup for the prototypes and create asset list	Limited to free assets
Week 6 13.07.2026	Prototype 1: Lighting Guidance	Scope becoming too large due to four separate prototypes Reuse assets and maintain a consistent level structure to combat this blocker
Week 7 20.07.2026	Prototype 2: Architectural Guidance	Possible time constraints
Week 8 27.07.2026	Prototype 3: Landmark/Audio Guidance	No Prior audio experience
Week 9 3.08.2026	Prototype 4: Combined Guidance Techniques	Possible time constraints

Week	Key milestone or deliverable	Potential blocker / contingency
Week 10 10.08.2026	Conduct user testing sessions and collect participant feedback through observation and questionnaires	Difficulty finding testers, ask classmates, volunteers or online participants
Week 11 17.08.2026	Continue to conduct user testing, start to analyse findings	
Week 12 24.08.2026	Analyse findings, compare results against literature, complete paper?	Insufficient data or unclear results to report upon

8. Process documentation

You are required to document your development process throughout the module. The method you choose should be appropriate to your artefact type and accessible to your assessor as part of your final submission. Your supervisor will confirm whether your proposed method is appropriate.

Proposed method

e.g. development blog, sketchbook (scanned/photographed), Git repository with commit history, design portfolio, production journal, annotated version control — or another method appropriate to your practice

Development blog / GitHub commit history

Link to GitHub Repo – <https://github.com/Mdot5596/MA-Dissertation>

Link to Blog – NA at the moment

Link to Devlogs - NA at the moment

Justification

Why is this method appropriate for your artefact type and working practice? How will it be made accessible within your final submission?

I plan on documenting my research and development via YouTube “Devlogs”. I chose this method as I will be able to articulate my work best through talking and I can visually show my progress at the same time. I have also had experience doing this as for my undergrad dissertation I did fortnightly devlogs through the progress of my game’s development.

Additionally, I will be using GitHub when I move from research to starting to create the artefacts. I will use this to show my commit frequency.

Supervisor feedback

Please provide feedback on each section of the proposal, including specific comment on the proposed process documentation method. This form should be completed at or following the first supervision session after the proposal presentation.

(04/06/2026) Dr Jeff Howard

I read your proposal, which has many good aspects. The research question seems well-defined, and the artefacts to be produced are ambitious (4 prototypes) but sensible. I do think that the primary form this research will take is a software product. You can write a written article if you want to, but I don’t think this

is required. Rather, I would recommend that you do a more **thorough lit review of the available articles on level design and player navigation** (bearing in mind the thousands of results that this query returned from Google Scholar). Add a few of the most relevant sources that you find to your proposal form, and keep investigating relevant literature and context as you work on your artefacts. I would then use this lit review and the contextual review from GDC Vault and other sources to inform your work, i.e. to help you to make artefacts that are at the cutting edge of contemporary practice and therefore able to help you to test the best practices available. From my perspective, the vlogs are an appropriate form of project documentation if your audience is indie developers, and these vlogs are sufficient as project documentation as long as you reference the articles and talks from your contextual review in relation to your own work. I wonder if it makes sense to do a fourth prototype combining all navigation techniques, since it would seem intuitively that a combination of lighting, space, and audio is almost certain to be more effective than any one of these strategies in isolation. Perhaps a sub-question could be *how* to most effectively combine these guidance techniques rather than *whether* to combine them.

Supervisor signature / confirmation:

Date: